BasketWise

Project Proposal: Smart Grocery Recommendations from Shopping Data

Course	CSE 426: Data Mining and Warehouse Lab
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Project Type	Individual (Solo) Project
Date	09 November 2026

1. Introduction

Retail stores record thousands of purchases every day, and hidden inside this data are buying patterns that can improve product placement, promotions, and recommendations. BasketWise applies the techniques taught in the Data Mining Lab to a real grocery transaction dataset in order to discover which products are frequently bought together and to turn those patterns into an item recommendation system.

2. Problem Statement

Raw transaction data is messy and cannot be mined directly. It contains inconsistent item names, duplicate records, and missing values, and it is stored one item per row instead of one basket per row. The problem is to clean and restructure this data, extract reliable association rules, and use them to recommend items that a customer is likely to buy next.

3. Objectives

1 Clean and prepare raw grocery transaction data using data preprocessing techniques.

2 Convert the data into customer baskets suitable for pattern mining.

3 Generate frequent itemsets and association rules using Apriori and FP-Growth, and compare the two algorithms.

4 Build a recommendation function that suggests items for a given item or a whole basket.

5 Present the results through a simple website built from the exported results.

4. Dataset

The project uses the Kaggle *Groceries Dataset*, which contains about 38,000 purchase records with three attributes: member number, date, and item description. One basket is defined as all items bought by one member on one date.

5. Methodology

Step	Description	Lab Topic
1	Explore data, handle missing and hidden missing values, remove duplicates, standardize item names, parse dates	Data Preprocessing
2	Group records into baskets and one-hot encode them	Data Preparation
3	Mine frequent itemsets with Apriori and FP-Growth and compare runtime and results	Association Rule Mining
4	Generate rules and evaluate them using support, confidence, and lift	Association Rule Mining
5	Build item-based and basket-based recommendation functions, with an optional similarity-based comparison	Recommendation System
6	Visualize rules and export results as JSON files	Visualization

6. Tools and Technologies

- **Notebook:** Google Colab, Python, pandas, NumPy, matplotlib, seaborn, mlxtend
- **Website:** Next.js, which displays the JSON results exported from the notebook

7. Expected Outcomes

- A fully documented Colab notebook covering preprocessing, mining, evaluation, and recommendation. This is the core deliverable of the project.
- A ranked set of association rules with clear interpretations.
- A working recommender that returns the top items for a selected item or basket.
- A companion BasketWise website that lets users search for an item and view its recommendations, the rules, and summary charts.

8. Conclusion

This project connects several lab topics into one practical workflow, from raw data to a usable recommendation system. It demonstrates how data mining can convert everyday transaction records into useful business insight.